Why a Single-Qubit Gate Can Become Expensive

A compact technical write-up on Clifford+T synthesis, injection, and STAR logical rotations

Team Synthesis + Team STAR | Bloqade/Squin + Tsim

Question and Scope

The challenge starts from a gate that looks harmless:

$$R_Z(\theta) = e^{-i\theta Z/2}, \qquad R_Z(\pi/2^n),\ n \in \{0, 1, 2, \ldots\}.$$

As an abstract operation this is a one-qubit rotation. The point of the project is to track what happens once the same operation is implemented under realistic constraints: a finite Clifford+T gate set, indirect non-Clifford gates, and logical error correction.

We used the challenge distance

$$d(U, V) = \sqrt{1 - \frac{|\mathrm{Tr}(U^\dagger V)|}{2}}, \tag{1}$$

which is zero when the target U and implementation V agree up to global phase. The main resource chain studied in the synthesis track is

$$R_Z(\pi/2^n) \longrightarrow \text{Clifford+}T \text{ sequence} \longrightarrow T\text{-injection gadgets} \longrightarrow \text{logical Steane implementation.}$$

The STAR track studies the same bottleneck from a different angle: can an arbitrary-angle logical rotation be implemented on a surface-code block without a full magic-state factory?

Team Synthesis: From Rotation to Resource

Direct Clifford+T synthesis

For $n = 0, 1, 2$, the first rotations are exact up to global phase:

$$R_Z(\pi) \sim Z = S^2, \qquad R_Z(\pi/2) \sim S, \qquad R_Z(\pi/4) \sim T.$$

For $n \geq 3$, we used Ross–Selinger synthesis through Qiskit's `gridsynth_rz`, which produces ancilla-free Clifford+T approximations of Z rotations [1]. The implementation pattern was simple: handle exact cases first, synthesize the remaining angles, extract the gate sequence, and load it as a one-qubit Squin kernel.

```
def Rz_gate(n, epsilon=1e-10):
    if n == 0: return X_gate       # Rz(pi), up to phase
    if n == 1: return S_kernel
    if n == 2: return T_kernel

    theta = np.pi / 2**n
    qk_circ = gridsynth_rz(theta, epsilon=epsilon)
    sequence = gate_sequence_from_circuit(qk_circ)
    return load_circuit_as_squin_kernel(sequence)
```

The relevant control parameter is not only n, but also the demanded synthesis precision ϵ. In our data, the T-count grows approximately linearly in $\log_2(1/\epsilon)$, with slope close to 3, as expected for Ross–Selinger synthesis.

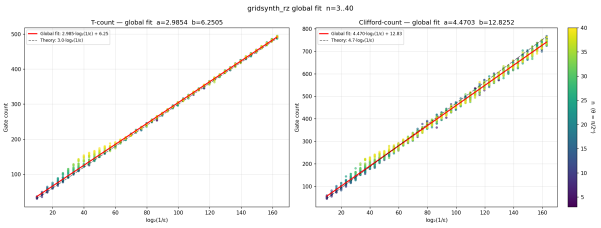

Figure 1: Observed Clifford+T scaling for direct synthesis.

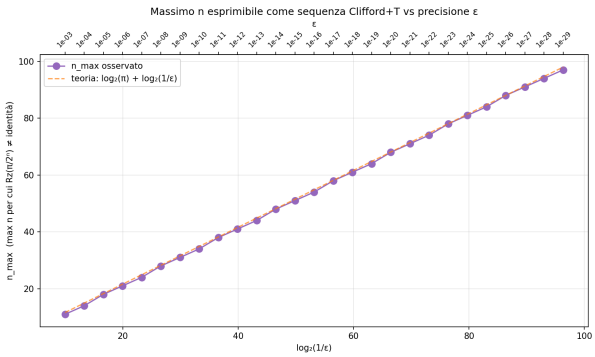

Figure 2: Largest distinguishable n as a function of ϵ.

A useful nuance is that extremely small rotations are close to the identity. For a fixed tolerance ϵ, there is therefore a largest useful n:

$$n_{\max} \approx \log_2(\pi) + \log_2(1/\epsilon). \tag{2}$$

Thus rotations can be made as small as desired, provided that ϵ is made small enough. The cost is that the Clifford+T sequence becomes longer.

Replacing direct T gates by injection

The next constraint was to assume that T cannot be applied directly to the data qubit. This is the fault-tolerant situation in miniature: Clifford gates are usually easier, while non-Clifford gates must be supplied through resource states. We used the standard magic-state injection picture [2]:

$$T \quad \longrightarrow \quad |A\rangle = T\,|+\rangle + \text{CNOT} + \text{measurement} + \text{feed-forward}.$$

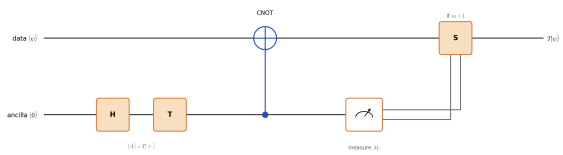

Figure 3: Magic-state injection circuit for one T gate.

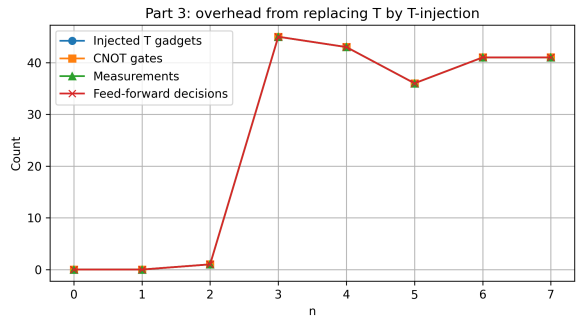

Figure 4: Injection overhead after expanding each $T/T^\dagger$.

Consequently, a Clifford+T word with m non-Clifford gates no longer costs just m one-qubit operations. It costs approximately m magic states, m CNOTs, m measurements, and m feed-forward decisions.

If feed-forward is unavailable, the correction branch has to be handled by postselection. With success probability 1/2 per injected gate, a sequence with m injected $T/T^\dagger$ gates succeeds with probability

$$p_{\text{success}} = 2^{-m}, \qquad \mathbb{E}[\text{trials}] = 2^m.$$

This is where a short-looking one-qubit rotation can become operationally expensive.

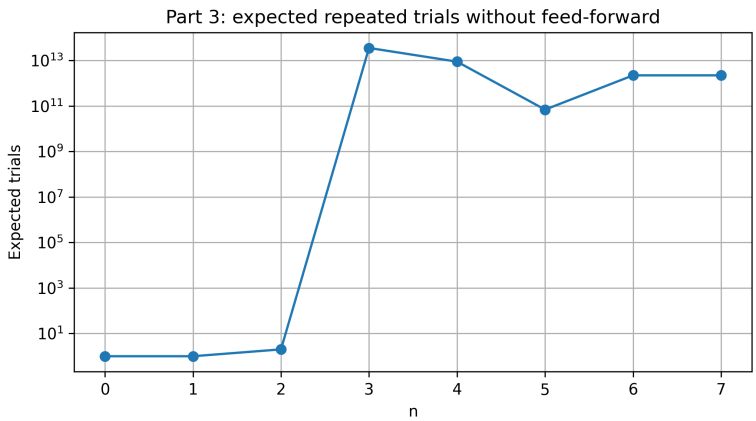

Figure 5: Expected repetitions when injected gates are implemented by postselection rather than active feed-forward.

Logical Steane implementation

The final synthesis step was to replace the physical qubit by one logical qubit encoded in the Steane $[[7, 1, 3]]$ code. In this code, logical H, S, and CNOT are transversal or comparatively natural, while T is not transversal. This is not accidental: the Eastin–Knill theorem forbids a universal set of transversal gates for a quantum error-correcting code [3].

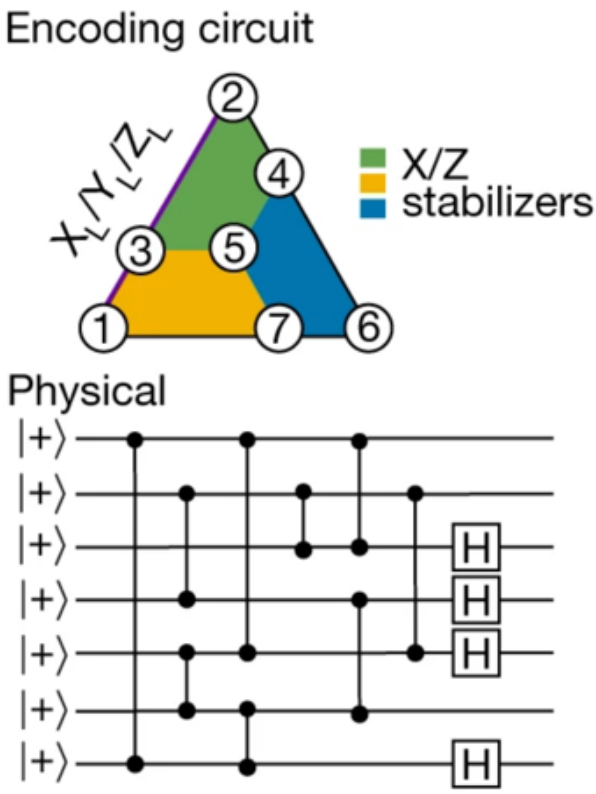

Figure 6: Steane-code circuit used to prepare $|0_L\rangle$ [5].

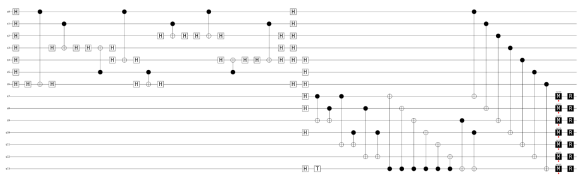

Figure 7: Logical CNOT implemented transversally between two blocks.

At the logical level, injection mirrors the physical gadget, but every object is encoded:

$$T_L \quad \longrightarrow \quad |A_L\rangle + \mathrm{CNOT}_L + \text{logical measurement} + \text{logical correction}.$$

One logical magic state already needs a seven-qubit block, and one logical CNOT is a layer of

seven physical CNOTs. Therefore the T-count from synthesis is really a count of logical injection gadgets.

Synthesis lesson

The important distinction is not "one-qubit" versus "two-qubit". It is *transversal* versus *non-transversal*. At the physical level, T is a one-qubit gate. At the logical fault-tolerant level, in codes such as Steane and the surface code, T is not transversal. Every synthesized $T/T^\dagger$ inside $R_Z(\pi/2^n)$ can therefore become encoded ancilla preparation, logical CNOTs, measurement, and feed-forward or postselection.

Team STAR: Logical Rotations on the Surface Code

Surface-code setup and modular Squin construction

The STAR track studies the rotated distance-d surface code for $d \in \{3, 5, 7\}$, corresponding to $d^2 = 9, 25, 49$ data qubits and $d^2 - 1 = 8, 24, 48$ syndrome ancillas. Error correction repeatedly measures Z stabilizers, which detect X errors, and X stabilizers, which detect Z errors. Increasing d suppresses logical errors, but increases both qubit count and circuit depth.

Rather than treating the reference Stim files as opaque circuits, we rebuilt the STAR circuits modularly in Squin. A single primitive, `build_stabilizer_supports(d)`, constructs the stabilizer layout. A factory then returns the three reusable kernels:

```
initialize_qubits, syndrome_round, physical_rotation = make_helpers(d, angle)
```

Here `initialize_qubits` projects into the code space, `syndrome_round` performs one full Z+X stabilizer cycle, and `physical_rotation` applies the transversal physical rotation to the first column. The noisy factory adds `squin.depolarize` and `squin.depolarize2` without changing the circuit structure.

STAR rotation under noise

STAR, Space-Time Analog Rotation, applies $R_Z(\theta_{\text{phys}})$ to the first column of d data qubits, then uses d syndrome rounds and a logical unrotation to distill a logical $R_Z(\theta_L)$ [4]. The angles are related by

$$\theta_{\text{phys}} = 2 \arctan\left(\tan\left(\frac{\theta_L}{2}\right)^{1/d}\right). \tag{3}$$

For small θ_L, this behaves as $\theta_{\text{phys}} \approx \theta_L^{1/d}$, so the physical rotation is larger than the target logical angle.

Our final noisy implementation used separate clean/noisy syndrome rounds, optional final-readout noise, and adaptive batching until a target number of accepted shots was reached. Postselection is essential: shots with detector firings in the post-rotation rounds are discarded. The logical error rate is then computed over accepted shots and normalized as p_L/d.

Teleporting the rotation

STAR can also prepare an arbitrary-angle logical resource state

$$|m_0\rangle = R_Z^L(\theta)\,|+\rangle_L\,,$$

which is then consumed by a teleportation circuit. Following Akahoshi et al., the noiseless protocol is:

```
|psi>_L  -- [CNOT ctrl] -- M_Z_L  ->  postselect on 0
|m0>_L   -- [CNOT tgt ] -- M_X_L  ->  R_Z(+/- theta)|psi>_L
```

Because Tsim does not support active feed-forward, we postselected on the branch $M_Z^L = 0$. The

implementation allocates independent main and STAR-ancilla code blocks, and the logical CNOT is simply a transversal loop:

```
for i in range(num_data):
    squin.cx(data_s[i], data_m[i])
```

This shows the advantage and limitation of STAR at once: it avoids a full magic state factory, but pays through syndrome rounds, postselection, and a logical error floor.

Combined Takeaway

Both tracks answer the same question. The synthesis track shows that precision turns a one-qubit rotation into a Clifford+T sequence, and fault tolerance turns every $T/T^\dagger$ into an encoded resource. The STAR track avoids long T sequences by using a partially fault-tolerant analog rotation protocol, but then pays in code distance, syndrome cycles, acceptance probability, and postselection.

Layer	**Main cost**
Direct synthesis	T-count grows with the demanded precision $\log(1/\epsilon)$.
T unavailable on data	Each $T/T^\dagger$ becomes magic state + CNOT + measurement + correction.
No feed-forward	Postselection gives success probability 2^{-m} for m injected gates.
Logical Steane	T is non-transversal, so each T becomes a logical injection gadget.
STAR surface-code route	Avoids a full magic factory, but needs syndrome rounds and postselection.

Table 1: Why the abstract arity of a gate does not determine its implementation cost.

The conclusion is compact: a gate can be mathematically one-qubit while being expensive at the fault-tolerant level. What matters is whether the operation is native, Clifford, transversal, and compatible with the architecture's correction mechanism.

References

[1] N. J. Ross and P. Selinger, *Optimal ancilla-free Clifford+T approximation of z-rotations*, Quantum Information & Computation, 2016. arXiv:1403.2975.

[2] S. Bravyi and A. Kitaev, *Universal quantum computation with ideal Clifford gates and noisy ancillas*, Physical Review A, 2005.

[3] B. Eastin and E. Knill, *Restrictions on transversal encoded quantum gate sets*, Physical Review Letters, 2009.

[4] Y. Akahoshi et al., *Partially Fault-Tolerant Quantum Computing Architecture with Error-Corrected Clifford Gates and Space-Time Efficient Analog Rotations*, 2023. arXiv:2303.13181.

[5] D. Bluvstein et al., *Logical quantum processor based on reconfigurable atom arrays*, Nature 626, 58–65, 2024. arXiv:2312.03982.